In this simulation, I explored Braitenberg Vehicles 3A and 3B, which build on the previous vehicle design by using combinations of sensor wiring (mono vs stereo), inhibition, crossing, and activation functions.

Behavior:

These vehicles simulate how different types of wiring and activation patterns affect movement in response to a light source (sun). The vehicle 3a has one sensor and it acts like vehicle 2 in behaviour. It goes close to the source and stays there admiring or showing in love behaviour whereas vehicle 3b has two sensors and acts like an explorer.

Logic:

I have tried to explain every step in the code file. Some of the important parts are:

- **Sensor Update:**
  Each vehicle updates its left and right sensor positions based on its current orientation. These sensors are slightly forward and offset (angled at 30 degrees) from the front of the vehicle body. Mono vehicles use only one sensor.
- **Activation Functions:**
  I tested three different ways to convert distance to light activation:
  inverse_distance: closer = higher activation
  threshold: only activates when the light is far away (like obstacle avoidance)
  sinoside: creates a wavy, fluctuating pattern of activation
  These functions shape how the vehicle reacts to the light based on sensor distance. In the code the inverse_distance and threshold has to be normalized to see the expected behaviour. Because all three functions have different ranges and can't be replaced as it is.
- **Inhibition and Wiring:**
  Each vehicle uses inhibitory wiring, meaning sensors reduce motor activity the closer they are to the sun. Additionally:
  3A (Mono): single sensor controls both motors equally.
  3B (Stereo + Crossed): left sensor controls right motor and vice versa.
- **Noise Toggle:**
  I added an optional noise toggle (press N to turn it on/off), which introduces random fluctuations in motor speed and turning direction, simulating real-world imperfection or neural noise.

Settings:

I kept most of the settings same as they were in vehicle 01:

Sun: radius=30, color=yellow, position=(600, 300)

Vehicle: radius=50
Vehicle 3A: red, mono, inhibitory, non-crossed
Vehicle 3B: blue, stereo, inhibitory, crossed wiring

Toggle: There is a toggle between mono/stereo using M key, and another one to turn on/off the added randomness using N key. These help observe how each parameter changes the path and reaction of the vehicle.

Sensor: The vehicle has two sensors, each positioned in front of it, that help it "see" the light source. These sensors are placed forward and at a slight offset from the vehicle's center.

Note:

The behavior of vehicles under threshold and inverse distance activations may not show expected results if the sensor activation values are too low or too subtle. Adjusting the scaling or threshold levels can help exaggerate the effect for clearer observations.